

the newly formed Chicago Research and Trading Group, make millions in the following weeks while other traders were left bewildered by the market's moves.

This moment marked one of the earliest applications of systematic data analysis in modern trading, a precursor to today's algorithmic approaches. Ginter didn't have Python or fancy visualization tools—just graph paper, a pencil, and a conviction that hidden patterns existed in market data if you knew how to look for them.

Today, we have infinitely more powerful tools at our fingertips. In the previous chapter, we learned how to collect and clean financial data. Now, we'll explore how to analyze and visualize this data to uncover actionable insights—the patterns and relationships that can drive profitable trading strategies.

Think of financial data as a dense forest. Without a map or compass, you could wander aimlessly, missing both dangers and opportunities. Data analysis and visualization provide that map—they help you navigate the complexity, spot potential trading signals, and avoid analytical dead ends. In this chapter, we'll explore two main approaches:

- 1.

Exploratory Data Analysis (EDA): Techniques to understand the basic characteristics of your financial data and identify potential patterns

- 2.

Time Series Analysis: Specialized methods for analyzing data that changes over time—the fundamental nature of financial markets

By the end of this chapter, you'll be equipped with powerful analytical tools that turn raw numbers into trading insights. Let's begin our exploration of how to see what others miss in the financial markets.

Exploratory Data Analysis: Getting to Know Your Data